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Laboratory Evaluation of Gas Dilution Systems for Analyzer Calibration and Calibration Gas Analysis

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Gas dilution systems can be used for multipoint calibrations of pollutant gas analyzers and for the analysis and certification of compressed gas calibration standards. This laboratory evaluation obtained estimates of the accuracy and precision of four representative gas dilution systems. Diluted and undiluted gas mixtures containing carbon monoxide, nitric oxide or sulfur dioxide in nitrogen were sampled by pollutant gas analyzers. Accuracies were estimated from the difference between the slopes of regression lines from measurements of the diluted and undiluted gas mixtures. These accuracy estimates ranged from -3.3 to 10.0 percent. Precisions were estimated from the 95-percent uncertainty for regression-predicted concentrations. These precision estimates ranged from 0.3 to 14.9 percent of the predicted concentration for mid-range dilutions.

Calibration gases often are diluted during the multipoint calibration of pollutant gas analyzers and during the analysis and certification of compressed gas calibration standards. Gas dilution systems are devices that allow two gas streams to be mixed together continuously and quantitatively. Such systems could be used for multipoint calibrations and analyses of gas mixtures. However, systematic and random errors associated with the dilution would be added to the total measurement uncertainty. Could one use gas dilution systems for these applications with acceptable levels of accuracy and precision?

Under U.S. Environmental Protection Agency (EPA) sponsorship, Research Triangle Institute (RTI) conducted a laboratory evaluation of four representative gas dilution systems. The goal of the evaluation was to estimate the accuracies and precisions of the systems using measurements by pollutant gas analyzers. These estimates were obtained from the statistical analysis of multipoint calibrations using undiluted and diluted gas mixtures.

Gas Dilution Systems

The EnviroNics Model 2020 continuous emissions monitoring calibration system uses thermal mass flow controllers to regulate the flow rates of a compressed gas calibration standard and dilution nitrogen, which are blended together. The system was configured to operate in both of the dilution regions that were used in the evaluation (i.e., 1 to 9 percent and 10 to 90 percent of the standard's concentration). Its specified flow rate accuracy is ± 0.5 percent of the full-scale range and its specified flow rate repeatability is ± 0.2 percent of the full-scale range.

The Milton Roy Model 821S gas divider uses 10 identical, parallel capillary tubes and solenoid valves to blend the calibration standard and the dilution nitrogen. The system was configured to operate in the 10 to 90 percent dilution region. Its specified dilution accuracy is ± 0.2 percent of the standard's concentration and its specified repeatability is ± 0.1 percent of the point.

The Wösthoff Model 1KM67 gas mixing pump uses three piston-driven, positive-displacement pumps to blend the calibration standard and the dilution nitrogen. The standard goes through

one pump and the nitrogen goes through the other two pumps. Switchable gears vary the pistons' stroke frequency and thereby vary the diluted gas concentration. The pumps and gears in the instrument used for this evaluation were lubricated with silicone oil. The system was configured to operate in the 1 to 9 percent dilution region. Its specified accuracy under calibration conditions is ± 1.0 percent relative to the selected dilution level.

The Wösthoff Model 5KA37 gas mixing pump uses five piston-driven, positive-displacement pumps to blend the calibration standard and the dilution nitrogen. The pumps are an oil-less type and have a fixed stroke frequency. Three stopcocks are used to switch the gas streams entering three of the five pumps and thereby vary the diluted gas concentration. The system was configured to operate in the 1 to 9 percent dilution region. Its specified accuracy under calibration conditions is ± 0.5 percent relative to the selected dilution level.

Experimental Procedures

Gas mixtures containing carbon monoxide (CO), nitric oxide (NO), or sulfur dioxide (SO₂) in nitrogen were measured during the evaluation. These measurements were made using the analytical instrumentation listed in Table I. Several gas dilution systems were compared over several days. All measurements that were obtained on the same day were considered to be comparable and no measurements were compared across different days.

Multipoint calibrations were performed using undiluted and diluted gas mixtures. The undiluted gas mixtures were National Institute of Standards and Technology Standard Reference Materials (NIST SRMs) and zero-grade nitrogen. These gas mixtures were the reference standards for the accuracy estimates. The diluted gas mixtures were obtained by dilution of EPA Protocol Gases, which are prepared and analyzed by specialty gas producers using an EPA-specified traceability protocol. RTI verified the certified concentrations of these standards relative to the concentrations of the SRMs.

The calibration data were analyzed using least-squares regression techniques under the assumption of a constant random error

term. For several multipoint calibrations, a quadratic equation fitted the data better than a straight-line equation. The accuracy of the gas dilution systems was estimated by comparing the slope of the regression line for the diluted gas mixtures with the corresponding slope for the undiluted gas mixtures.

$$\text{Accuracy} = 100 \left[\frac{\text{Diluted Slope} - \text{Undiluted Slope}}{\text{Undiluted Slope}} \right]$$

This value is constant across the range of predicted concentrations. Statistical tests were performed to check if the diluted slope was significantly different from the undiluted slope.

The precision of the gas dilution systems was estimated by the relative uncertainty of concentrations that are predicted from the regression equation.¹

$$\text{Precision} = 100 \left[\frac{95\% \text{ Uncertainty for Predicted Concentration}}{\text{Predicted Concentration}} \right]$$

Although the 95-percent uncertainty is approximately constant across the range of predicted concentrations, the precision esti-

mate varies inversely with the predicted concentration. This variation is illustrated in Table II which gives precision estimates obtained from the multipoint calibration of the NO analyzer on its 0 to 250 parts per million (ppm) range. The precision at the upper end of the regression line is approximately 0.5 percent, but the precision at the lower end is approximately nine times larger. In this paper, precision estimates are generally listed for dilutions of 5 or 50 percent of the diluted standard's concentration.

The precision estimates also vary with the number of measurements in the multipoint calibration and with the number of measurements of the sample being analyzed. This variation is illustrated in Table III, which gives precision estimates obtained from various subsets of the 250 ppm NO multipoint calibration. The precision estimates for a 5-point calibration and one sample measurement are approximately two to three times larger than the precision estimate for a 10-point calibration and three sample measurements. This example demonstrates that precision estimates for a gas dilution system can be improved by increasing the number of measurements.

Several unexpected events occurred during the evaluation. Excessive zero drift in the NO analyzer was discovered and repaired. Several days' data were discarded because of this zero drift. One of the capillaries in the Milton Roy 821S system became plugged during the evaluation and this system could no longer be used. The slope of the diluted SO₂ regression line for the Wösthoff 1KM67 system was significantly different from the slopes of the undiluted gas mixtures and the other systems' diluted gas mixtures. This difference may be due to interference of silicone oil from the Wösthoff 1KM67 system on the ultraviolet fluorescence SO₂ analyzer. Additionally, oil was seen coming from the system's nitrogen pump and a dropout trap was installed to prevent downstream oil contamination. Worse than expected precision estimates for the Wösthoff 5KA37 system prompted its inspection and repair by its manufacturer. A minor leak was discovered and the system was reevaluated at RTI after it was repaired.

Results

Several caveats concerning the evaluation should be considered as the results are reviewed. The evaluation studied only a limited range of experimental conditions. The evaluation did not attempt to obtain optimum performance (e.g., better precision estimates) from the gas dilution systems due to time constraints on the number of measurements that could be made during a single day. Not all of the gas dilution systems were evaluated in the same dilution region. The gas dilution systems that were evaluated were selected based only on their availability and their dilution principles. The evaluation did not assess the corresponding accuracy and precision of undiluted gas mixtures which might be used in multipoint calibrations.

The results of the evaluation are given in Table IV for the Environics and Milton

Table I. Analytical instrumentation for laboratory evaluation of gas dilution systems.

Gas Mixture	Pollutant Gas Analyzer	Analyzer Range (ppm)	
		For Dilution of 10% to 90% of Std. Conc.	For Dilution of 1 to 9% of Std. Conc.
CO/N ₂	Bendix 8501-5CA	1000	50
NO/N ₂	Thermo Electron 10AR	250	25 and 100
	Columbia Scientific 1600	—	5
SO ₂ /N ₂	Thermo Electron 40	1000	50

Table II. Comparison of precision estimates associated with different predicted concentrations for the same multipoint calibration.

NO Analyzer		Precision (percent) for One Sample Measurement		
Range (PPM)	Gas Dilution System Conc.	At Dilution of 90% of Std. Conc.	At Dilution of 50% of Std. Conc.	At Dilution of 20% of Std. Conc.
250	Envionics 2020	0.59	0.97	5.14
	Milton Roy 821S	0.50	0.85	4.53

Table III. Comparison of precision estimates associated with various number of measurements of calibration standards and sample.^{a,b}

Measured Values		Precision at Dilution of 50% of Std. Conc.	
Used in the Regression Calculations	No. of Values in Regression	1 Sample Measurement	3 Sample Measurements
Odd Values ^c	5	1.85	1.24
Even Values ^d	5	1.20	0.81
All Values	10	0.97	0.61
^a Gas Dilution System	Envionics 2020		
^b Analyzer Range	250 ppm NO		
^c Odd Values	10, 30, 50, 70, 90% of Std. Conc.		
^d Even Values	0, 20, 40, 60, 80% of Std. Conc.		

Roy systems at the higher-concentration dilution range (i.e., 10 to 90 percent of the standard concentration). The results for the SO₂ and CO multipoint calibrations are given for both straight-line and quadratic regressions. The SO₂ analyzer is slightly nonlinear and the CO analyzer is quite nonlinear. Accuracy estimates were not calculated for the quadratic regressions. The Milton Roy 821S system was not evaluated beyond the NO and SO₂ calibrations because of the plugged capillary.

The results of the evaluation are given in Table V for the Environics and Wösthoff systems at the lower-concentration dilution range (i.e., 1 to 9 percent of the standard concentration).

Table IV. Results for higher concentration dilution range (10 to 90 percent of standard concentration).

Gas Mixture	Analyzer Range (PPM)	Gas Dilution System	Slope Accuracy (Percent)	Precision at Dilution of 50% of Std. Conc.
NO/N ₂	250	Envionics 2020	1.60	0.97
		Milton Roy 821S	0.25	0.85
SO ₂ /N ₂	500	Envionics 2020	2.50 ^a	1.24
		Milton Roy 821S	0.74	2.38
SO ₂ /N ₂ ^b	500	Envionics 2020	—	0.76
		Milton Roy 821S	—	2.12
CO/N ₂	1000	Envionics 2020	-4.53 ^a	10.90
CO ₂ /N ₂ ^b	1000	Envionics 2020	—	0.47

^aStatistically significant difference between diluted and undiluted slopes

^bQuadratic regression

Table V. Results for lower concentration dilution range (1 to 9 percent of standard concentration).

Gas Mixture	Analyzer Range (PPM)	Gas Dilution System	Slope Accuracy (Percent)	Precision at Dilution of 5% of Std. Conc.
NO/N ₂	100	Envionics 2020	-0.72	1.01
		Wösthoff 1KM67	-0.53	1.05
		Wösthoff 5KA37	-0.43	14.48
NO/N ₂ (after repair)	100	Wösthoff 5KA37	1.25 ^a	1.52
		Wösthoff 5KA37	0.98 ^a	1.39
		Wösthoff 5KA37	1.08 ^a	2.13
		Wösthoff 5KA37	2.17 ^a	1.38
		Wösthoff 5KA37	2.11 ^a	1.59
NO/N ₂ ^b	5	Envionics 2020	1.39 ^a	0.31
		Wösthoff 1KM67	0.11	0.59
		Wösthoff 5KA37	0.34	12.64
SO ₂ /N ₂	50	Envionics 2020	2.62 ^a	3.13
		Wösthoff 1KM67 ^c	9.87 ^a	3.81
		Wösthoff 5KA37	0.51	14.91
SO ₂ /N ₂	50	Envionics 2020	-1.97 ^a	2.08
		Wösthoff 1KM67 ^c	10.04 ^a	5.22
		Wösthoff 5KA37	-1.52	12.28
CO/N ₂	50	Envionics 2020	-3.33 ^a	0.74
		Wösthoff 1KM67 ^c	-0.90 ^a	1.11
		Wösthoff 5KA37	1.09	9.75

^aStatistically significant difference between diluted and undiluted slopes

^bReference standard was diluted for those measurements

^cPossible interference effect from silicone oil

Results are given for the Wösthoff 5KA37 system both before and after its repair. The last two after-repair accuracy estimates may be biased due to a concentration shift in the SRM associated with low cylinder pressure. The SO₂ calibration was replicated to verify the slope accuracy estimate for the Wösthoff 1KM67 system.

Conclusions

The number of experiments conducted in the evaluation is too small to allow one to draw any definitive conclusions concerning the accuracy of the gas dilution systems. For 12 of the 25 accuracy estimates, there was no statistically significant difference between the slopes of the regression lines

for the undiluted and diluted gas mixtures. The values of the remaining 13 accuracy estimates ranged from -4.5 to 10.0 percent although the extreme values are associated with a nonlinear calibration curve and a possible interference effect. The accuracy estimates exhibit some variations from one day to the next and from one experimental condition to the next. In general, the accuracy estimates for the four gas dilution systems were similar.

The precision estimates for the four gas dilution systems were also generally similar. The values range from 0.3 to 14.9 percent for predicted concentrations in the middle portion of the regression line. The largest values are associated with a leaking system. The precision estimates exhibited some variations from one day to the next and from one experimental condition to the next. The precision estimates were significantly different in different portions of the regression line. Better precision can be obtained by making more measurements during multipoint calibrations and sample measurements.

Disclaimer

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